Doomscroll — Lab 3 Design Model (refined from Lab 2)

Solid border = BUILT in current code. Dashed border = PLANNED (not yet implemented). Shaded fill = new since Lab 2.

Boundary

«boundary»
AppScreen [PLANNED]

SetupScreen [PLANNED]

CalibrationScreen [PLANNED]

SettingsScreen [PLANNED]

ReaderOverlay [BUILT, simplified]
cv2 debug window today, not a real HUD

RecalibrationPromptDialog [PLANNED]

WebcamInterface [BUILT]

ScreenCaptureInterface [PLANNED]

OSScrollAPI [BUILT]

FaceLandmarkBackend [BUILT, new]
wraps MediaPipe face-mesh (478 pts/frame)

WindowBoundsProvider [BUILT, new]
OS focused-window geometry (Win32/Cocoa)

Control

CalibrationController [BUILT, split]

GazeEstimationController [BUILT, refactored]
owns HeadPoseEstimator + GazePointSmoother,
delegates prediction to a Strategy:

«interface» GazePredictionStrategy

LinearCalibration Strategy

MLModel Strategy

GazePointSmoother (Kalman + EMA, innovation-gated)

ContentAnalysisController [PLANNED]

AutoscrollController [BUILT, inline]
dwell-trigger logic today, not a standalone class yet

SettingsController [PLANNED]

DatasetBuilderController [BUILT, new]
recording -> labeled feature/label arrays

ModelTrainingController [BUILT, new]
compares ridge/mlp/hgb, selects + persists best

Entity

User [PLANNED: no persistence]

CalibrationProfile [BUILT, two forms]
linear-calib params, or ml/model.py AffineCorrection

AutoscrollSettings [PLANNED]

Session [BUILT, repurposed]
= one recorded dataset, not "one app usage period"

HeadPose [BUILT]

GazeVector [BUILT]

GazePoint [BUILT]

ContentElement [PLANNED]

ImportantContentRegion [PLANNED]

ScrollCommand [BUILT, implicit]
not an explicit object yet (direct injector calls)

FeatureVector [BUILT, new]
11-dim: gaze offsets, lid ratio, yaw/pitch/roll, head xyz

TrainedGazeModel [BUILT, new]
persisted estimator + metadata + optional correction

LabeledFrames / SessionDataset [BUILT, new]
LabelSettings: lag_ms, idle-filter, click-mode

BUILT, carried over from Lab 2 (solid border)

BUILT, new since Lab 2 (heavier border)

PLANNED, not yet implemented (dashed border)

association / dependency

generalization / interface realization

Full class-by-class rationale: lab3/design_model.md. Unchanged Lab 2 relationships (e.g. User 1..1 CalibrationProfile) are omitted here for legibility -- see conceptual_model.pdf.